

Statistical Mechanics notes

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1 Lesson 1, 23/09

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2 Lesson 2, 29/09

2.1 Internal Energy as a TD potential

[10 min. cfr with notes pt 2 on moodle/Remi] $N = \text{const.}$

2.2 Legendre Transform

What if the control variables are the intensive ones, e.g. system in a thermostat? It is useful to use T instead of S : the mathematical tool used is the *Legendre transform* (the one used to transform $L(q, \dot{q}) \rightarrow H(q, p)$, using $p = \frac{\partial L}{\partial \dot{q}}$)

$$E(S, V, N, X) \rightarrow F(T, V, N, X) := E(S, V, N, X) - TS$$

where S is found by inverting the relation $T = \frac{\partial E}{\partial S}$. F is the Helmholtz free energy

Using the fundamental equation of TD

$$E = TS - pV + \mu N + yX$$

the function F can also be written as

$$F = -pV + \mu N + yX :$$

differentiating this yields

$$dF \leq -SdT - pdV + \mu dN + ydX$$

so keeping fixed T, V, N, X , the equilibrium is at minimal F .

What are the conjugated variables? As always

$$\begin{aligned} S &= -\frac{\partial F}{\partial T} \\ p &= -\frac{\partial F}{\partial V} \\ \mu &= \frac{\partial F}{\partial N} \end{aligned}$$

This process can be done with every intensive variable e.g. enthalpy:

$$E(S, V, N, X) \xrightarrow[p = -\frac{\partial E}{\partial V}]{} H(S, p, N, X) := E(S, V, N, X) + pV$$

and

$$H = TS + \mu N + yX \xrightarrow[\text{d and fund. eq.}]{} dH \leq TdS + vdp + \mu dN + ydX$$

so fixing S, p, N, X , the equilibrium is at minimal H .

Doing this with respect to both couples (T, S) and (p, V) yields the Gibbs free energy:

$$G(T, p, N, X) = E(S, V, N, X) - TS + pV$$

using the fundamental equation yields

$$G = \mu N + yX$$

Note In simple systems (without (y, X)) $G = \mu N$. The differential is

$$dG \leq -SdT + Vdp + \mu dN + ydX$$

that shows that, fixing T, p, N, X , equilibrium is at minimal G .

The last Legendre Transform of interest is the one that yields the *Grand Canonical potential*:

$$E(S, V, N, X) \xrightarrow{(T, S), (\mu, N)} \bar{\omega}(T, V, \mu, X) = E(S, V, N, X) - TS - \mu N$$

The fundamental equation yields

$$\bar{\omega} = -pV + yX$$

and differentiating yields

$$d\bar{\omega} \leq -SdT - pdV - Nd\mu + ydX$$

so the equilibrium is at the minimum of $\bar{\omega}$.

The pressure is simply

$$p = -\frac{\partial \bar{\omega}}{\partial V} = -\frac{\bar{\omega}}{V}$$

2.3 Response Functions

These functions describe the reaction of variables when another one changes (typically conjugated variables), are very easy to measure and they are deeply connected to the size of the fluctuations of the system (Fluctuation-dissipation theorem).

Def. Heat capacity

$$C = \frac{\delta Q}{dT}$$

is the amount of heat to be furnished to change the T ; it is not an exact differential, so it depends on the transformation.

Heat capacity at constant (V, N) : if $dV = dN = dX = 0 \rightarrow \delta W = 0$; by the first principle

$$\delta Q = dE$$

so

$$C_{V, N, X} = C_V = \frac{\delta Q}{dT} = \left. \frac{\partial E}{\partial T} \right|_{V, N, X}$$

or, using the definition of entropy

$$\delta Q = TdS$$

and fixing (V, N, X)

$$dS = \left. \frac{\partial S}{\partial T} \right|_{V, N, X} dT$$

so

$$C_V = \frac{TdS}{dT} = T \left(\frac{\partial S}{\partial T} \right)_{V, N, X}.$$

A more natural approach springs from the Helmholtz free energy:

$$S = - \left. \frac{\partial F}{\partial T} \right|_{V, N, X} \rightarrow C_V = -T \frac{\partial^2 F}{\partial T^2}$$

Heat capacity at constant (p, N, X) : the first principle states

$$\delta W = pdV \rightarrow \delta Q = dE + pdV$$

so

$$C_p = \frac{\delta Q}{dT} = \left. \frac{\partial E}{\partial T} \right|_{p,V,X} + p \left. \frac{\partial V}{\partial T} \right|_{p,V,X}$$

but also

$$E = G + TS - pV = G - T \frac{\partial G}{\partial T} - pV$$

so

$$C_p = \frac{\partial}{\partial T} \left(G - T \frac{\partial G}{\partial T} - pV \right) + p \frac{\partial V}{\partial T} = -T \left. \frac{\partial^2 G}{\partial T^2} \right|_{p,N,X}$$

2.4 Mechanical response functions

Susceptibility χ is the volume response to a change in pressure: this also varies depending on the transformation.

Isothermal: fixing (T, N, X)

$$\chi_T := - \left. \frac{\partial V}{\partial p} \right|_{T,N,X}$$

since the variables are (T, p, N, X) , with only p independent, the most natural TD potential to use is G : some calculations show that

$$\chi_T = - \left. \frac{\partial^2 G}{\partial p^2} \right|_{T,N,X}$$

Adiabatic: fixing (S, N, X)

$$\chi_S = - \left. \frac{\partial V}{\partial p} \right|_{S,N,X}$$

the TD potential is the enthalpy H :

$$V = \left. \frac{\partial H}{\partial p} \right|_{S,N,X} \rightarrow \chi_S = \left. \frac{\partial^2 H}{\partial p^2} \right|_{S,N,X}$$

All these response functions are between conjugated variables: what happens if they are not? Consider the Thermal expansion coefficient

$$\alpha_{p,N,X} = \alpha_p = \left. \frac{\partial V}{\partial T} \right|_{p,N,X} \stackrel{\text{exercise}}{=} \left. \frac{\partial^2 G}{\partial p \partial T} \right|_{p,N,X}$$

2.5 Maxwell Relations

These are relations between TD quantities: consider the set of independent variables

$$\{Z_A\}$$

their conjugated variables

$$\{\xi_A\}$$

and a TD potential

$$d\Phi = \sum \xi_A dZ_A$$

it follows that

$$\frac{\partial \xi_A}{\partial Z_B} = \frac{\partial^2 \Phi}{\partial Z_A \partial Z_B} = \frac{\partial^2 \Phi}{\partial Z_B \partial Z_A} = \frac{\partial \xi_B}{\partial Z_A}$$

from other mathematical properties (like the "chain rule", see Billò notes) arise other relations. One finds

$$\frac{1}{T} C_p > \frac{1}{T} C_V + \frac{\alpha_p^2}{\chi_T}$$

2.6 Stability of equilibrium state

Even at equilibrium, at maximum S , there can be small fluctuations in TD variables. For all such fluctuations, the entropy **must decrease** to ensure stability.

What are the conditions for local equilibrium? Consider an isolated system at equilibrium subdivided in 2 parts, which can exchange particles and heat.

$$\begin{aligned}E &= E_1 + E_2 \\V &= V_1 + V_2 \\N &= N_1 + N_2\end{aligned}$$

Suppose some fluctuation occurs in E_i, V_i, N_i : the other part of the system must have an opposite fluctuation

$$\Delta E_2 = -\Delta E_1 \dots$$

Assuming small fluctuations, the variation of S at first order is

$$\Delta S = \sum_{i=1}^2 \left(\left. \frac{\partial S_i}{\partial E_i} \right|_{\text{eq}} \Delta E_i + \left. \frac{\partial S_i}{\partial V_i} \right|_{\text{eq}} \Delta V_i + \left. \frac{\partial S_i}{\partial N_i} \right|_{\text{eq}} \Delta N_i \right) = \left(\frac{1}{T_1^{\text{eq}}} - \frac{1}{T_2^{\text{eq}}} \right) \Delta E_1 + \left(\frac{p_1^{\text{eq}}}{T_1^{\text{eq}}} - \frac{p_2^{\text{eq}}}{T_2^{\text{eq}}} \right) \Delta V_1 + \dots$$

Since $\Delta S \leq 0$, it must be that all the quantities are equal

$$T_1^{\text{eq}} = T_2^{\text{eq}} \dots$$

Iterating this process for smaller and smaller partitions of the system proves that, unsurprisingly, at first order

$$\Delta S = 0.$$

The chemical potential condition

$$\mu_1^{\text{eq}} = \mu_2^{\text{eq}}$$

guarantees chemical equilibrium: supposing an isolated system with $\delta Q = \delta W = 0$, $\Delta S > 0$ and

$$\Delta S \approx - \left(\frac{\mu_1}{T_1} - \frac{\mu_2}{T} \right) \Delta N_1 > 0$$

so if $\mu_1 > \mu_2$ it must be that

$$\Delta N_1 < 0,$$

in other words, matter flows from higher to lower μ .

The quantity

$$-\nabla \left(\frac{\mu}{T} \right)$$

is a generalized force (see first lesson), much like $\nabla \left(\frac{1}{T} \right)$.

3 Lesson 3, 30/09

3.1 Equilibrium in external field

The TD system is often subject to external forces, like gravity: how can equilibrium be described in this case?

Consider a TD system and subdivide it into smaller systems s.t they are still TD systems with a large number of particles: the coordinates of the portion is $\mathbf{x}$.

Since μ is the energy needed to add particles to the system, the control variables used are T, p, N : from the definition of G

$$\mu = \frac{G}{N}$$

and from the fundamental equations

$$\frac{\partial \mu}{\partial p} = \frac{1}{N} \frac{\partial G}{\partial p} := \frac{V(T, p, N)}{N} = v(T, p)$$

where v is the volume per particle.

The chemical potential consists of two parts

$$\mu(T, p; \mathbf{x}) = \mu_0(T, p) + u(\mathbf{x})$$

where μ_0 is μ with the external field turned off and u is related to the field.

At the equilibrium, one expects that T remains constant, p depends on $\mathbf{x}$ (like the Stevino law) and μ remains constant and homogeneous.

These conditions translate to

$$\frac{\partial \mu}{\partial x^i} = 0 = \frac{\partial \mu}{\partial p} \cdot \frac{\partial p}{\partial x^i} + \frac{\partial u}{\partial x^i}$$

so

$$v(T, p) \frac{\partial p}{\partial x^i} + \frac{\partial u}{\partial x^i} = 0$$

Example: gravity In this case $u(\mathbf{x}) = mgz$, so

$$\frac{dp}{dz} v(T, p) = -mg$$

so v does not depend on T , adding the incompressibility of the fluid imposes constant v , so

$$\frac{dp}{dz} = -\frac{m}{v}gz = -\rho gz \rightarrow p(z) = p_0 - \rho gz$$

in the case of an ideal gas, using $p v = k_b T$ yields (prove it as an exercise)

$$p(T, z) = p_0 \exp \left[-\frac{mg}{kT} z \right]$$

3.2 Second order equilibrium

It was found that, at first order

$$T, p, \mu \text{ are homogeneous} \\ \Delta S = 0$$

To discuss the second order fluctuations, it is needed to re-express everything in terms of TD potentials: consider the control variables

$$\left\{ \underbrace{X_1, \dots, X_r}_{\text{extensive}}; \underbrace{I_{r+1}, \dots, I_n}_{\text{intensive}} \right\}$$

and the potential

$$\Phi(X_i; I_j) :$$

its differential is

$$d\Phi = \sum_{i=1}^r I_i dX_i - \sum_{j=r+1}^n X_j dI_j$$

Note This generalizes the problem to have r extensive control variables and $r - n$ (?) intensive control variables.

Consider a TD system subdivided in two parts and a fluctuation in the extensive variables:

$$\delta X_i = \delta X_{i,1} + \delta X_{i,2} = 0$$

at first order

$$\Delta\Phi^{(1)} = \sum_{i=1}^r \left[\left(\frac{\partial\Phi}{\partial X_i} \right)_1 - \left(\frac{\partial\Phi}{\partial X_i} \right)_2 \right] \delta X_{i,1} \geq 0$$

As before, the only way to ensure this condition is

$$\left(\frac{\partial\Phi}{\partial X_i} \right)_1 = \left(\frac{\partial\Phi}{\partial X_i} \right)_2 \leftrightarrow I_{i,1} = I_{i,2}$$

with finer partitions this means that the intensive variables conjugated to the extensive control variables are homogeneous in the system, at least at first order.

To get a non-trivial result, consider the second order of fluctuation of a single variable X_i (i.e. considering the fluctuations independent):

$$\Delta\Phi^{(2)} = \frac{1}{2} \left[\left(\frac{\partial^2\Phi}{\partial X_i^2} \right)_1 + \left(\frac{\partial^2\Phi}{\partial X_i^2} \right)_2 \right] (\delta X_{i,1})^2 \geq 0$$

where the minus sign was destroyed by the square; this statement is true only if

$$\frac{\partial^2\Phi}{\partial X_i^2} = \frac{\partial I_i}{\partial X_i} \geq 0$$

from this follows that

$$\frac{\partial X_i}{\partial I_i} \geq 0$$

and, remembering the Legendre transform

$$\Phi(..., X_i, ...) \rightarrow \tilde{\Phi}(..., I_i, ...) \leftrightarrow X_i = -\frac{\partial\tilde{\Phi}}{\partial I_i}$$

this gives

$$\frac{\partial^2\tilde{\Phi}}{\partial I_i^2} \leq 0$$

This gives a geometric condition on the TD potentials at equilibrium: they are either a concave function of intensive variables or convex functions of extensive variables (or vice-versa **CHECK!!!!!!**) **Example** Consider the Helmholtz free energy: the stability condition implies that

$$\frac{\partial^2 F}{\partial T^2} \leq 0$$

but

$$C_V = -T \frac{\partial^2 F}{\partial T^2} \rightarrow C_V \geq 0$$

and also

$$\underbrace{\frac{\partial^2 F}{\partial V^2}}_{\geq 0} = -\frac{\partial p}{\partial V} = -\frac{1}{\frac{\partial V}{\partial p}} = \frac{1}{\chi_T} \geq 0 \leftrightarrow \chi_T \geq 0$$

Example Consider the Gibbs energy $G(T, p, N)$:

$$\frac{\partial^2 G}{\partial T^2} \leq 0$$

and the C_p definition

$$C_p = -T \frac{\partial^2 G}{\partial T^2} \geq 0.$$

Again the χ_T definition is

$$\chi_T = - \underbrace{\frac{\partial^2 G}{\partial p^2}}_{\leq 0} \geq 0$$

which is consistent with the previous example.

From the relation

$$C_p = C_V + \frac{T \alpha_p^2}{\chi_T}$$

and the stability conditions discussed it follows that, in general

$$C_p \geq C_V$$

3.3 Multi-component systems

Some TD systems have different components mixed (e.g. the atmosphere), indexed with $a = 1, 2 \dots c$ so the chemical potential is not unique: μ_a, N_a and these potentials depend separately from those variables (which vars???)

$$dE = TdS - pdV + \sum_{a=1}^c \mu_a dN_a + ydX$$

Each potential has its own equilibrium condition $\mu_{a,1} = \mu_{a,2}$ and intensive variables depend only on $c - 1$ ratios

$$\nu_a = \frac{N_a}{N}$$

with the constraint

$$\sum_{a=1}^c N_a = N \leftrightarrow \sum_{a=1}^c \nu_a = 1$$

3.4 Phase transitions

This topic is very difficult to treat mathematically and one of the most important results of statistical mechanics is the description of this phenomenon.

What are the conditions of coexistence of different phases of matter in equilibrium (e.g. ice and water)?

At equilibrium, the state of the TD system is characterized by E, V, N : it is not necessary that at equilibrium state these variables are homogeneous.

Each different phase (each defined by its state equation) of the system has its set of variables p_i, T_i, μ_i .

If the state is equilibrium, the intensive variables must be homogeneous:

$$p_1 = p_2, \quad T_1 = T_2, \quad \mu_1 = \mu_2$$

it is then natural to take p, T as control variables, making μ a dependent variable; since the phases have different equations of state, the potentials will be different: for the phase α

$$G_\alpha(T, p, N_\alpha) = \mu_\alpha N_\alpha \rightarrow \mu_\alpha = \frac{G_\alpha}{N_\alpha}.$$

The phases can only coexist on *curves of coexistence* in the state space: for generic values of p, T , $\mu_\alpha \neq \mu_\beta$ and so the phases do not coexist and, considering N constant, the system "selects" the phase which minimizes

$$\mu = \frac{G}{N}.$$

So phases α and β coexist if and only if the values of p, T are so that

$$\mu_\alpha(p, T) = \mu_\beta(p, T)$$

which typically is a curve $p(T)$ in the state space. When this curve is crossed, a phase transition occurs and it can be of two natures.

- **First order transition**

The coexistence curve is determined by the intersection of the surfaces $\mu_\alpha(T, p)$ and $\mu_\beta(T, p)$. In general, the value of the derivatives differ at the intersection point(s): fixing p they are

$$\left(\frac{\partial \mu_\alpha}{\partial T}\right)_p = \frac{1}{N_\alpha} \left(\frac{\partial G}{\partial T}\right)_p = -\frac{S_\alpha}{N_\alpha} := -s_\alpha$$

which is the entropy per particle, so in general

$$s_\alpha \neq s_\beta$$

Suppose $s_\alpha > s_\beta$: for $T = T(p) - \varepsilon$, $S_{(-)} = Ns_\beta$, while for $T = T(p) + \varepsilon$, $S_{(+)} = Ns_\alpha$; in this case the transition from β to α makes the entropy jump of

$$\Delta S = N(s_\alpha - s_\beta) > 0$$

this is nothing but the effect of latent heat, the heat to be furnished to the system just to do a phase transition

$$L = T(p)\Delta S.$$

From an enthalpy p.o.v.,

$$H(S, p, N) = G + TS$$

since $\mu_\alpha = \mu_\beta$, $\Delta G = 0$ and so

$$\Delta H = \Delta(TS) = T\Delta S = L.$$

This is similar for fixed temperature transitions, with derivative

$$\left(\frac{\partial \mu}{\partial p}\right)_{T, N} = \frac{V}{N} = v$$

which is the volume per particle; so in general $v_\alpha \neq v_\beta$ and the volume of the system changes during the transition *even at constant pressure* (e.g. ice melting).